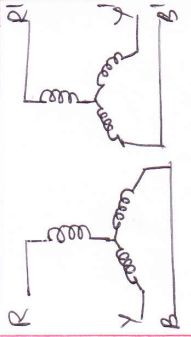
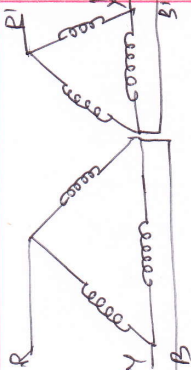
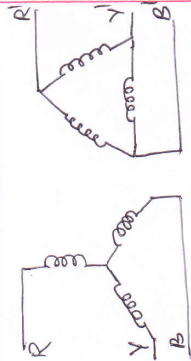
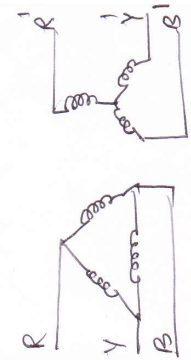


Let, line voltage at 1st side = V line current at 1st side = I line voltage at 2nd side = V' line current at 2nd side = I' phase turns ratio $\frac{N_1}{N_2} = \alpha$	Star - star ($\Delta - \Delta$) 	Delta - Delta ($\Delta - \Delta$) 	Star - Delta ($\Delta - \Delta$) 	Delta - Star ($\Delta - \Delta$) 
Phase voltage at primary side	$V_1 = \frac{V}{\sqrt{3}}$	$V_1 = V$	$V_1 = \frac{V}{\sqrt{3}}$	$V_1 = V$
$\frac{V_1}{V_2} = \frac{N_1}{N_2} = \alpha = \frac{I_2}{I_1}$ $V_2 = \frac{V_1}{\alpha}$ $\therefore$ Phase voltage at secondary side	$V_2 = \frac{V_1}{\alpha} = \frac{V}{\alpha\sqrt{3}}$	$V_2 = \frac{V}{\alpha}$	$V_2 = \frac{V_1}{\alpha} = \frac{V}{\alpha\sqrt{3}}$	$V_2 = \frac{V_1}{\alpha} = \frac{V}{\alpha}$
line voltage at 2nd side	$V' = \sqrt{3} V_2 = \frac{V}{\alpha}$	$V' = V_2 = \frac{V}{\alpha}$	$V' = V_2 = \frac{V}{\alpha\sqrt{3}}$	$V' = \sqrt{3} V_2 = \frac{\sqrt{3}V}{\alpha}$
Phase current at 1st side	$I_1 = I$	$I_1 = I/\sqrt{3}$	$I_1 = I$	$I_1 = I/\sqrt{3}$
$\frac{I_2}{I_1} = \alpha \Rightarrow I_2 = \alpha I_1$ Phase current at 2nd	$I_2 = \alpha I_1 = \alpha I$	$I_2 = \alpha I_1 = \frac{\alpha I}{\sqrt{3}}$	$I_2 = \alpha I_1 = \alpha I$	$I_2 = \alpha I_1 = \frac{\alpha I}{\sqrt{3}}$
line current at 2nd	$I' = I_2 = \alpha I$	$I' = I_2 \sqrt{3} = \alpha I$	$I' = \sqrt{3} I_2 = \alpha \sqrt{3} I$	$I' = I_2 = \frac{\alpha I}{\sqrt{3}}$